Analysis of the Effects of Housing Permits on Population Growth from 2010-2022: Neural Networks

Introduction

Population growth is one of the most important factors reflecting how communities change over time, as it can affect (and be affected by) labor market trends, infrastructure quality, public services, and demand for housing, making it crucial to understand the factors that have the strongest effects on population growth. One of the most well studied variables is the availability of housing, as the availability of housing can not only affect the price of housing (thereby affecting the cost of living) but also the physical capacity for a city to hold more people and hire more people within the area itself. House permits, which are a very strong correlate of housing construction, give a clear indication of changes in housing supply, and can be used to examine the relationship between construction and population growth. Additionally, house permitting is affected by a multitude of factors, ranging from the amount of land left for zoning, to the regulatory environment in the area. However, population growth is affected by many socioeconomic, demographic, and labor market factors, making the relationship between permitting and population growth far too complicated for simple statistical modeling.

The goal of this project is to evaluate the relationship between house permitting (a strong correlate of house construction) and population growth across U.S. counties from 2010-2022. Using county-level housing, socioeconomic, demographic, and labor market data, this project applies multiple statistical approaches, and machine learning methods to determine the relationship from different perspectives, including linear relationships, nonlinear prediction methods, growth forecasting, and causal inference. This specific report focuses on neural

network techniques to determine whether a highly flexible machine learning model can improve population growth predictions. The model is designed to capture complex nonlinear relationships between permitting and county-level characteristics, while also determining whether strong predictive performance can be obtained with a relatively small number of neurons, rather than relying on a needlessly complex neural network.

Data

In order to determine the effect of housing permits on population growth, there were many covariates that were used for different reasons. These included the total permit rate per 1,000 people (obtained from the total number of permits in the county, divided by the population, multiplied by 1,000), as well as the number of single family permits per 1,000, the number of multifamily permits per 1,000 (specifically 2-4 unit permits), and the number of permits for buildings with at least 5 units (per 1,000). Each permit-rate variable was used as the main independent variable for its own regression, while the total permit rate was used for every other type of model. Additionally, the remaining housing variables included county-level homeownership rates, the growth rate in occupied housing units (which was lagged by a year), rent burden (the percentage of people paying more than 30% of their income in rent), and median house value. The labor market and economic characteristics that were used included median household income and the employment population ratio. The demographic variables that were used were the percent of people in a county who are white, the percent of people in a county who are black, the percent of people born outside the US, the change in the percent of people born outside the US (essentially functioning as net migration), as well as the county's median age. Lastly, I also used population density for the counties, which functions as a ceiling on how many

people can live in an area before geographic constraints inhibit housing construction (albeit, for many counties, that limit hasn't been reached). After finding the correct specifications for all of the variables, I also added two interaction terms, permits multiplied by housing growth, and permits multiplied by median house value.

Data Preparation

With the exception of the permit rate, I gathered all of the data from the American Community Survey's 5 year estimates. I used 5 year estimates instead of 1 year estimates because ACS data becomes very noisy when applied to small counties due to their small sample sizes for data collection. For the permit rate, I gathered information on total permits from the Department of Housing and Urban Development, which has the data from 1980-2022, after which I divided the number of each permit type by the population, and I multiplied them by 1,000 to get the per 1,000 rate. I used data from 2010-2022 for almost all of the variables because much of the ACS data starts in 2010, while the housing permit data set runs up to 2022. I also used 2023 data for population, foreign-born population, and the number of occupied housing units. These were used to create growth rates which were mostly used for lags so that I could use the 2022 data to determine trends. The data for the variables (from ACS data) was collected using API keys, and they were merged using SQL joins (mostly inner joins), with some variable transformations and calculations being performed in SQL using window functions, with other transformations being performed either in Excel or in Python workbooks.

For all of the models, I had to make some decisions about what data to include and what to remove. Firstly, since the COVID-19 pandemic falls within the time frame of this project, there are some years where population growth is significantly lower as a result of a crisis that

can’t easily be seen in the data. The period when this issue was most apparent was 2020-2021, as population growth was significantly lowered from 2020 to 2021 as hundreds of thousands died from the virus. While even more people died in 2021, this ended up having less of a deleterious effect on population growth because of the substantial increases in immigration during 2021. Since corresponding population growth was the variable of choice, population growth from 2020 to 2021 was associated with the values for each of the variables in 2020. As a result of the issues that 2020 would present for the data, I made the decision to remove the 2020 data from consideration, as well as its corresponding population growth. Although this made models like forecasting far more difficult, the alternative was worse as a whole year’s worth of data would be of incredibly poor quality.

Additionally, since there were some very noticeable outliers which could heavily alter the relationship between an independent and dependent variable, I tended towards removing them. This was firstly done by removing counties with a population below 2,500 as population growth figures are incredibly noisy at such low levels, since even a couple families deciding to move out can have a substantial effect on population growth. Additional outliers which were removed included counties which allowed more than 20 permits per 1,000 (a sample of roughly 30 observations), counties which had population growth rates above 8%, and below -5%, (a sample of roughly 40 observations), and counties which saw their number of available housing units either increase by more than 15% in one year, or decrease by more than 12.5% in one year (a sample of roughly 160 observations). While these removed a meaningful amount of data (roughly 0.6% of the data in the sample), even small amounts of outliers were capable of changing the best relationship between an IV and a DV from a linear relationship to a

logarithmic one, which is especially problematic for something like regression modeling, where the wrong relationship being specified can make models perform far worse.

Variables

Besides the house permit rate, which is the main independent variable for the project, the housing variables in question were used to communicate how affordable housing is in some way, or to communicate the availability of housing in full. Median house value, rent burden, and homeownership rate are used for different reasons to tell how affordable housing is in terms of the price of housing, the household level constraints from owning housing, or how many people own homes (because many people who don't own houses would own them if they could). The growth in housing units is a reflection of the housing stock at any given time, and even though it could be highly correlated with permitting frequency, it isn't, even when it's lagged.

The employment population ratio was used because it's a simple reflection of how many people work and is superior to the employment rate in some respects because it can better account for an area having lots of discouraged workers, retirees, or limited employment opportunities. Median household income was used to determine how wealthy the average person is in an area.

When it came to demographic variables, I used the three groups (white people, black people, and foreign born people) for different reasons. There is a well documented history of white people moving out of urban areas and into suburban areas when they become more racially diverse, especially with respect to black people moving into an area, and this pattern has been present as suburbs have become more diverse as well (even in middle class neighborhoods), so a higher share of black people in an area may mean that white people move to a nearby

county[1][2][3]. I used the white percentage to convey the level of racial homogeneity which may be a correlate of increased housing values for the same reason that an increased presence of minorities may be a correlate of decreased house values. I used the foreign born share not only as a variable related to diversity, but also as a way to show a crude form of net migration which is itself a massive component of population growth that may not be captured by the data in any other way, especially relevant because the reason for immigrants coming to a certain county aren't likely to be properly predicted from the data. While immigrant flows are somewhat correlated with local labor market conditions, the nature of a labor market can only partially be inferred from the data in question[4].

Lastly, I used population density to communicate the extent to which a county has already been urbanized, its ability to get more permits approved, and its capacity to become more densely populated. I chose this as a measure instead of an urban vs rural percentage because the definition of an urban area has become more restrictive with time, including through an increase in the minimum population requirement.

Methodology

In order to make the proper models, it's critical to ensure that each variable is correctly specified. This is done by checking for the shape of each variable when plotted against population growth, along with the R^2 and the RMSE (root mean squared error) having a minor role in determining whether a transformation is best. However, if the relationship between a given variable and population growth looks linear, I defaulted to using a linear transformation to avoid the risk of fixating too much on small deviations in R^2 and RMSE. Since population growth is more or less normally distributed, it does not require a logarithmic transformation.

Therefore, the other variables are measured against a dependent variable which has a linear specification. Ultimately, for most variables, a linear specification was adequate. This includes permits per 1,000 (which applies to all 4 of the permit variables), homeownership rate, the employment population ratio, the change in occupied housing units, the change in the percent of foreign born people in a county, the rent burden, the median house value, and the median age in years, as well as the two interaction terms which used linear specifications for both of their variables. Logarithmic specifications were used for the foreign born population share, as well as population density. A quadratic specification was used for the percent of people in a county who are white, and a cubic specification was used for the percent of people in a county who are black, while a reciprocal specification was used for the median household income.

A neural network is a machine learning model based on how our brains process information[5]. They have interconnected layers of neurons which find patterns in data and use them to make predictions. They consist of an input layer, hidden layers, and output layers. Neural networks have major components like neurons, connections, weights, biases, propagation functions, and learning rules, among others. The neurons are the fundamental units which receive inputs, and each neuron has a threshold and an activation function which helps to incorporate nonlinearity in the analysis, making the network capable of learning more complex patterns. Connections are what link to neurons and carry information from one neuron to the next, while being bound by weights and biases. Weights and biases are used to ascertain the strength and influence of connections between neurons. Propagation functions are used to help process data and transfer it across the layers of neurons. Lastly, the learning rule is what adjusts weights and biases to improve the network's accuracy.

In neural networks, learning comes from a three stage process, with input computation, output generation, and iterative refinement. Input computation is when data is given to the neural network. Output generation is when the network takes the given parameters and uses them to give an output. Afterwards, iterative refinement is when the network adjusts weights and biases in order to improve predictive performance for an array of tasks. Neural networks are used to determine and understand highly complex patterns, learn from data while improving model performance with continuous retraining and refinement, along with being used to power major technologies like natural language processing and automated systems and functioning as the fundamental foundation for AI.

Neural networks have three layers, those being the input layer, the hidden layer (which can have multiple layers), and the output layer. The input layer receives the input data, with each input neuron representing one feature/independent variable used for the network. The hidden layers perform most of the work, with each layer having a number of neurons that help transform the inputs into something that the output layer can receive, with it being extremely important to attain optimal predictive performance while avoiding overfitting. The output layer gives the output of the model and the output depends on the number of dependent variables or whether the model is classification or regression (the latter of which is the case for this project).

Neural networks start with forward propagation, where data is put into the network and passes from the input layer to the hidden layers to the output layer. During forward propagation, each neuron receives inputs multiplied by weights based on their connections, with the products being added together to form a sum, which has a bias added to it. Subsequently, the linear output goes through an activation function which adds a non-linear component into the network,

allowing it to learn complex patterns. For the purposes of this project, ReLU was the activation function.

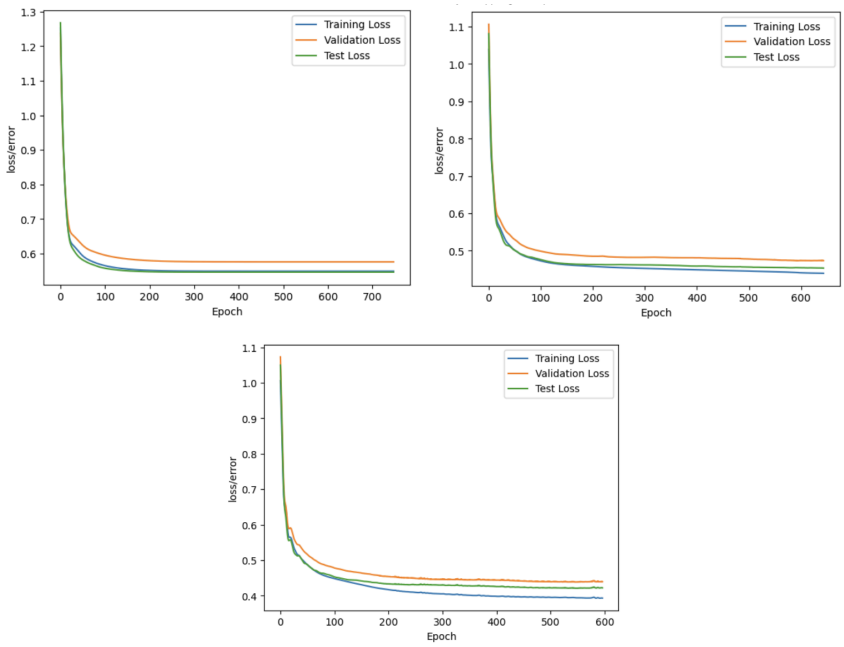
Following forward propagation, the neural network undergoes back propagation, starting with a loss function to measure performance. Loss functions measure the difference between actual and predicted outputs, with lower losses being better. Loss functions use loss calculations which determine the amount of error in the network predictions. Loss functions can be evaluated using a variety of function types. For this project, I used mean squared error which is the value of residuals squared and subsequently averaged. Afterwards, gradient calculation is performed where a neural network evaluates gradients for loss functions with respect to weights and biases, using the chain rule to determine how much of the output error is a result of each weight and bias. Lastly, weight updates are performed to optimize weights and biases, with the weights adjusted in the opposite direction of the gradients to minimize losses. Finally, the process is repeated across many iterations, which ultimately reduce losses and make the neural network's predictions more accurate.

Results

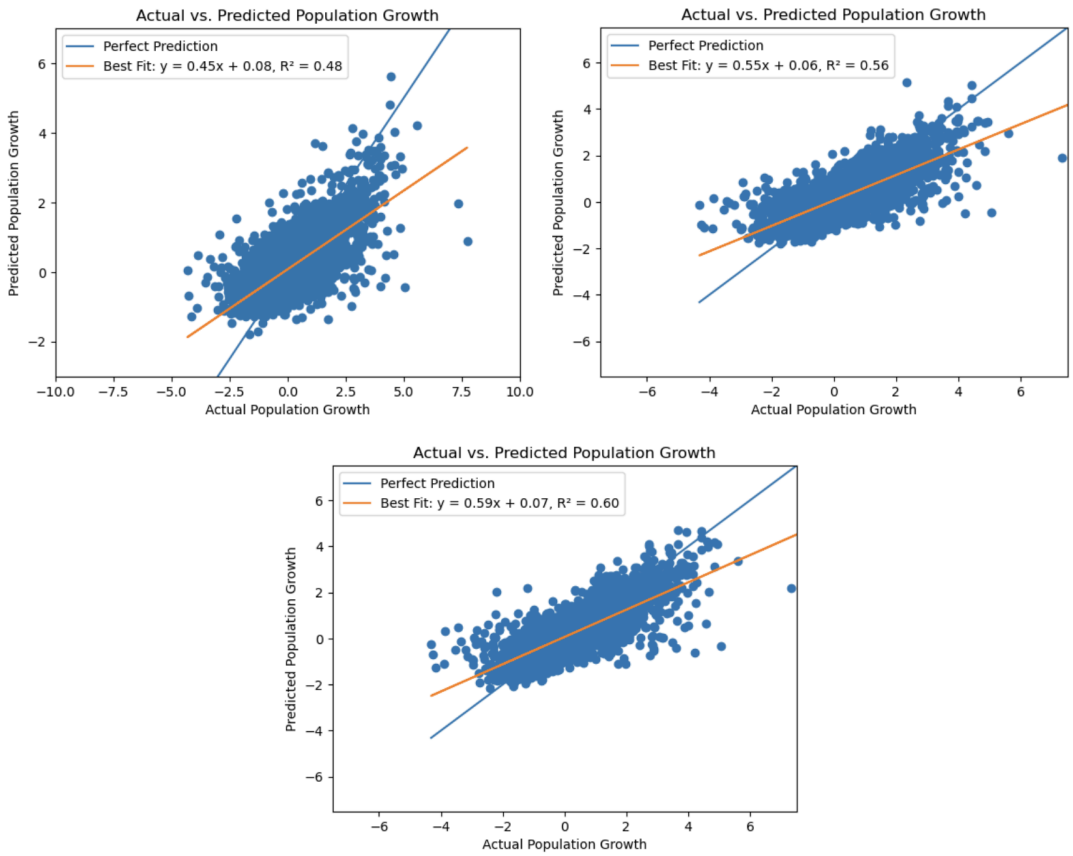
In order to find the optimal number of neurons, I used two approaches. I compared the test loss for a neural network based on the data with all variables being linear (which had zero hidden layers, thereby functioning like a regression) to one with the transformations that I found were best when analyzing the data for regressions. From there, I determined that the best model with one hidden layer was one with 10 neurons, while the best model with two hidden layers was one with 16 neurons in the first layer, and 12 in the second layer. The model with no hidden layers (only evaluating linear specifications of the variables) had a best training loss of 0.5494, a

best validation loss of 0.5764, and a best test loss of 0.5467, and an R^2 of 0.48, with the trendline between the actual and expected values of population growth being 0.45.

On the other hand, the model with one hidden layer had a best training loss of 0.4393, a best validation loss of 0.4738, and a best test loss of 0.4534 and an R^2 of 0.56, with a slope of 0.55 between the actual and expected values of population growth. The model with two hidden layers had a best training loss of 0.3933, a best validation loss of 0.4402, and a best test loss of 0.4227, with an R^2 of 0.60, and a slope of 0.59 between the actual and expected values of population growth. In spite of the increase in model complexity, the neural network with two hidden layers didn't suffer from any real signs of overfitting. While the gap between the training and test losses was larger at 0.03, vs 0.014 in the model with one hidden layer, the gap still wasn't particularly large. The graphs of the loss functions for the three models are shown in figures 1-3. After obtaining the loss functions, the actual vs predicted plots for population growth were made, with the results shown below in figures 4-6:



Figures 1-3: Loss Functions For the Neural Networks With Zero, One, and Two Hidden Layers Respectively.



Figures 4-6: Actual vs Predicted Population Growth Plots For the Neural Networks With Zero, One, and Two Hidden Layers Respectively.

Discussion

The neural network found that the optimal models were those with hidden layers modeling data with the necessary transformations, such as logs, reciprocals, and quadratic specifications when needed as shown from the better results based on loss functions, R^2 and actual vs predicted performance with respect to population growth, meaning that a neural network based solely on linear specifications is a worse model. The neural network with two hidden layers performed slightly better than the one with one hidden layer, however, the difference in performance wasn't substantially better with a test loss of 0.4227 vs 0.4534 in spite

of the added model complexity. However, even after the added model complexity, the neural network with multiple hidden layers didn't show any major signs of overfitting, and both neural networks have their strengths. The neural networks with hidden layers performed better than the regression models, however, they performed worse than the baseline gradient boosting model, and the model using a year shuffle split.

Crucially, this model, and many other prediction models in the project suffered from a major issue, which was that the variables used for this project simply didn't cover enough of the variation in population growth in order to be able to accurately predict actual population growth, and as a result, most attempts to plot actual population growth vs expected population growth ended up with slope coefficients that were roughly equal to the R^2 of the model. In practice, it is acceptable to have a model with an R^2 of 0.5 or below, but a slope of 0.5 is problematic in this context. What that means is that when the actual population growth is 1%, it expects 0.5% on average, when expected growth is -3%, it expects -1.5%, etc., but in practice, the slope should be 1. While it's expected for the data to have a decent amount of variance, as population growth is a very noisy variable that doesn't necessarily reflect a place being a great place to live, the model is hindered by its inability to more accurately determine growth. In order to improve the predictive performance of the model, some additions to this project could include a model with variables related to crime rate, public services, tax burden, climate, and school quality, among others.

Appendix

The housing permits dataset was obtained from the department of housing and urban development[6]. The ACS data included the datasets for Selected Social Characteristics, Selected

Housing Characteristics, Income in the Past 12 Months, Place of Birth by Poverty Status in the Past 12 Months in the United States, Age and Sex, Hispanic or Latino Origin by Race, Selected Economic Characteristics, and the Annual Geographic Information Table (GEOINFO)[7][8][9][10][11][12][13][14]. The code for the neural network was based on a video series for neural networks with PyTorch, with the code published on github[15].

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